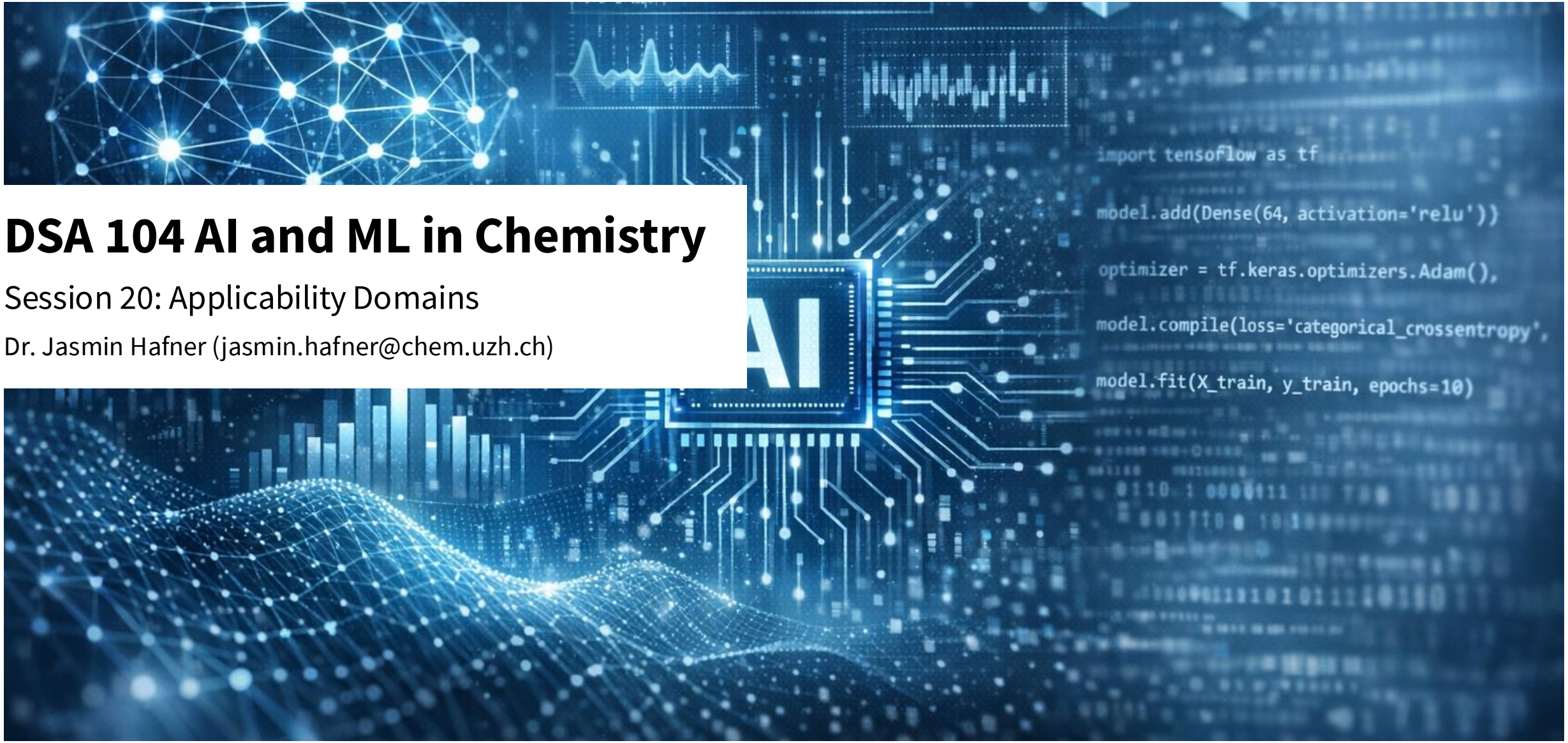
Universität Zürich[UZH]

Department of Chemistry



DSA 104 AI and ML in Chemistry

Session 20: Applicability Domains

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Lesson outline

Introduction: When (not) to trust a model

Definition: Applicability Domain (AD)

Types of ADs

Applications in chemistry: Molecular similarity

Discussion: When (not) to trust a model

When would you **NOT** expect a correct answer from your favourite LLM (ChatGPT, Copilot, …) :

- Meaning of an expression in your local dialect?
- The name of your best friend's dog (that you unfortunately forgot)?
- The structure of a molecule?
- Whether your roommate ate your ice cream in the freezer while you were away?
- The solution of the upcoming assignment ?

How do you know when you can trust a model?

Exercise: When (not) to trust a model

1) Build a model to predict logP on the ESOL data (e.g., a Random Forest using MACCS features)

2) Come up with a set of test molecules: A hormone, a PFAS, an endocrine disruptor, a metal composition, …

3) Use your model to predict logP for the test molecules

4) Can you trust the predictions? Discuss

5) Plot the training data features in 2-D (PCA, UMAP, …) and add the test molecules in a different color

(Q)SARs: (Quantitative) Structure-Activity Relationships

QSAR: A model that derives a substance property/activity from molecular structure

→ **Key assumption**: similar substances have similar properties/activities

12 | QSAR is "the Oldest" AI

Machine intelligence, which is normally presented as AI, refers to the intelligence exhibited by computers. QSAR modeling is the precursor of the modern AI or better IT IS a modeling method of AI. Already in 1995, Combes and Judson [192] commented on structure-based approaches for predicting toxicity end points, as QSARs, defining them as AI systems.

In [23], a paper written by some of the most famous QSAR modelers, it is also written: "... one can say that AI has been used in chemistry since the 1960's under the name QSAR."

Gramatica, P. (2025), Origin of the OECD Principles for QSAR Validation and Their Role in Changing the QSAR Paradigm Worldwide: An Historical Overview. Journal of Chemometrics, 39: e70014. https://doi.org/10.1002/cem.70014

Regulatory QSARs: When we NEED to trust a model

- **QSARs** can be used in **REACH** (Registration Evaluation Authorization of Chemicals) to ***predict*** substance **properties**, as an alternative to resource-intensive animal / in vivo / environmental testing

- QSAR for **regulatory purpose:**

  "[…] a (Q)SAR should be associated with a **defined domain of applicability**"

  - In-domain → use normally
  - Marginal → caution / consult other models
  - Out-of-domain → no prediction, request data

- **Regulatory AD definition** in the guidelines for the *(Q)SAR Model Reporting Format (QMRF)* :

  "APPLICABILITY DOMAIN refers to the response and chemical structure space in which the model makes predictions with a given reliability. Ideally the applicability domain should express the structural, physicochemical and response space of the model. The CHEMICAL STRUCTURE (x variable) space can be expressed by information on physicochemical properties and/or structural fragments. The RESPONSE (y variable) can be any physicochemical, biological or environmental effect that is being predicted."

https://qsardb.org/guidelines/qmrf
Guidelines: https://qsardb.org/guidelines/qmrf/QMRF_v1.2-2008-05-26.pdf

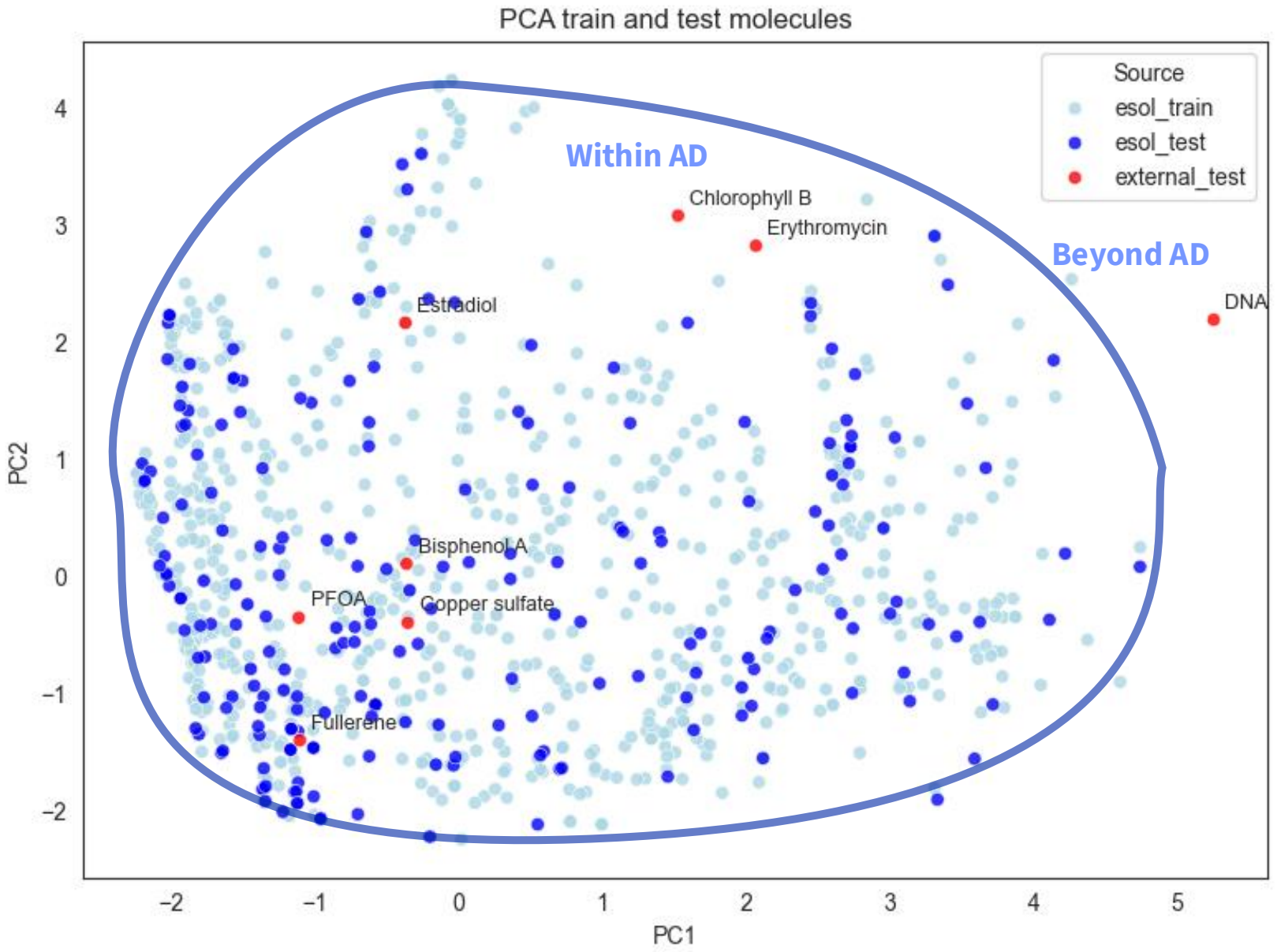
Applicability Domain (AD)

- The AD is the region of the **descriptor space** where a model's predictions are expected to be **accurate and reliable**

- **Outside** of AD → model **performance degradation**

- Particularly important for **property** or **activity** predictions of molecules or materials

- In general: ML models are good at **interpolation** within training data, but not so good at **extrapolation**

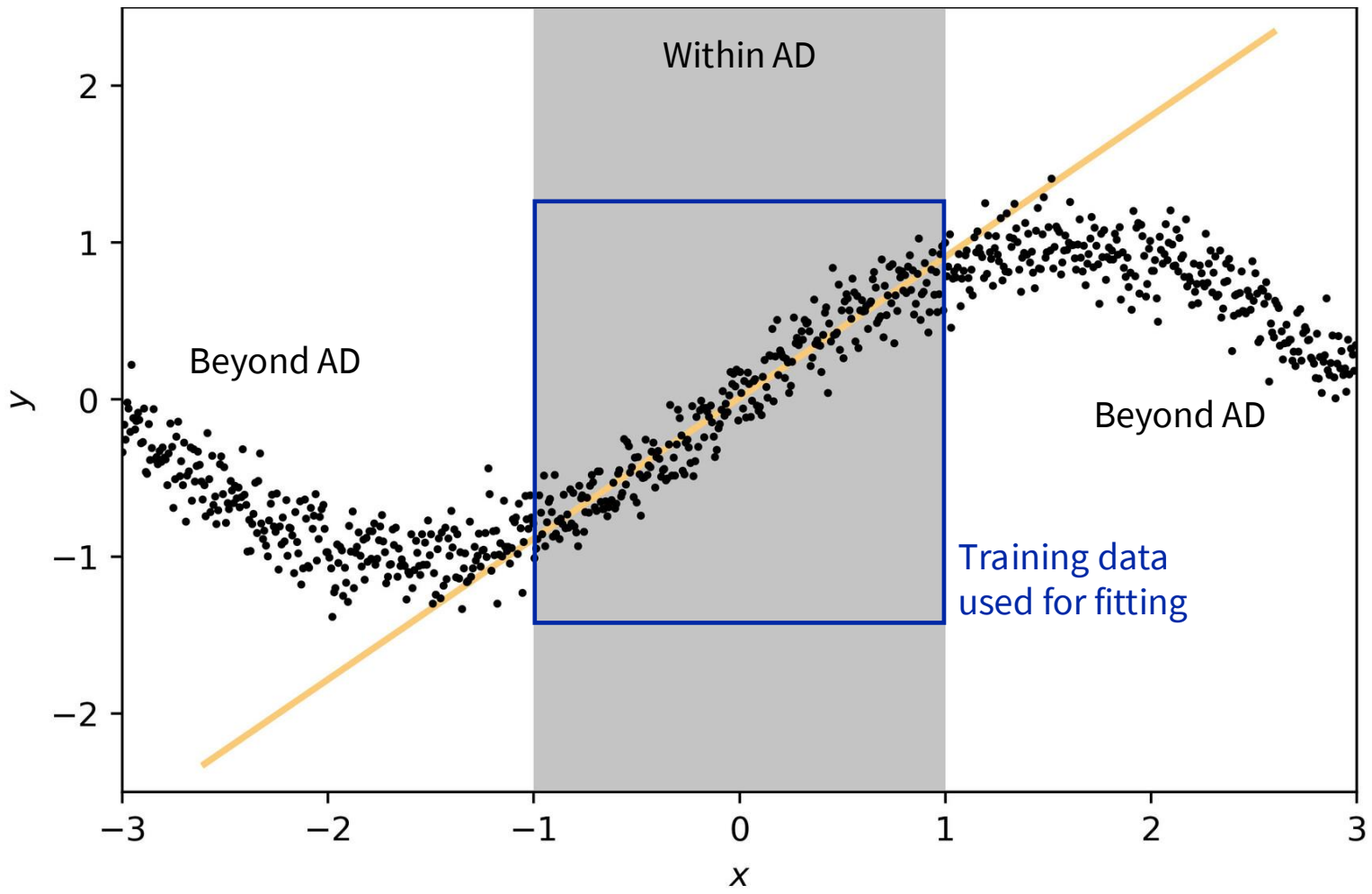


Schultz, L.E., Wang, Y., Jacobs, R. *et al.* A general approach for determining applicability domain of machine learning models. *npj Comput Mater* **11**, 95 (2025). https://doi.org/10.1038/s41524-025-01573-x

AD metrics

1. Feature / target variable range
2. Leverage
3. Kernel Density Estimation
4. Distance to k-Nearest Neighbors
5. Chemical similarity (Tanimoto coefficient)

AD metrics - Feature range



x: feature (here: 1-dimensional)

y: response variable

yellow line: fitted model

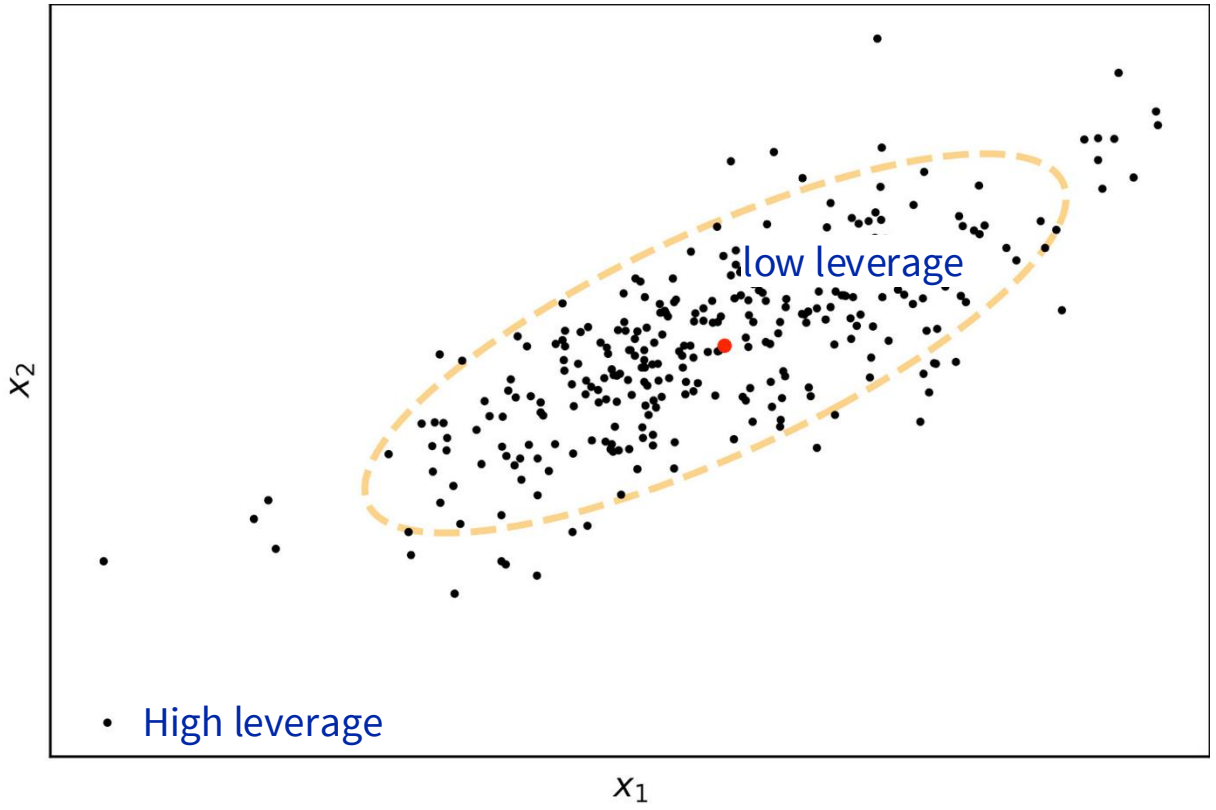
Question:
How would you define the AD of this model?

Within AD: **-1 < x <1**

T.-M.Dutschmann, V.Schlenker, K.Baumann, Molecular Informatics2024, 43, e202400018. https://doi.org/10.1002/minf.202400018

AD metrics – Leverage / Mahalanobis Distance from centroid

Leverage: measure of the effect of a particular observation on model outcome due to the position of that observation in the space of the inputs.



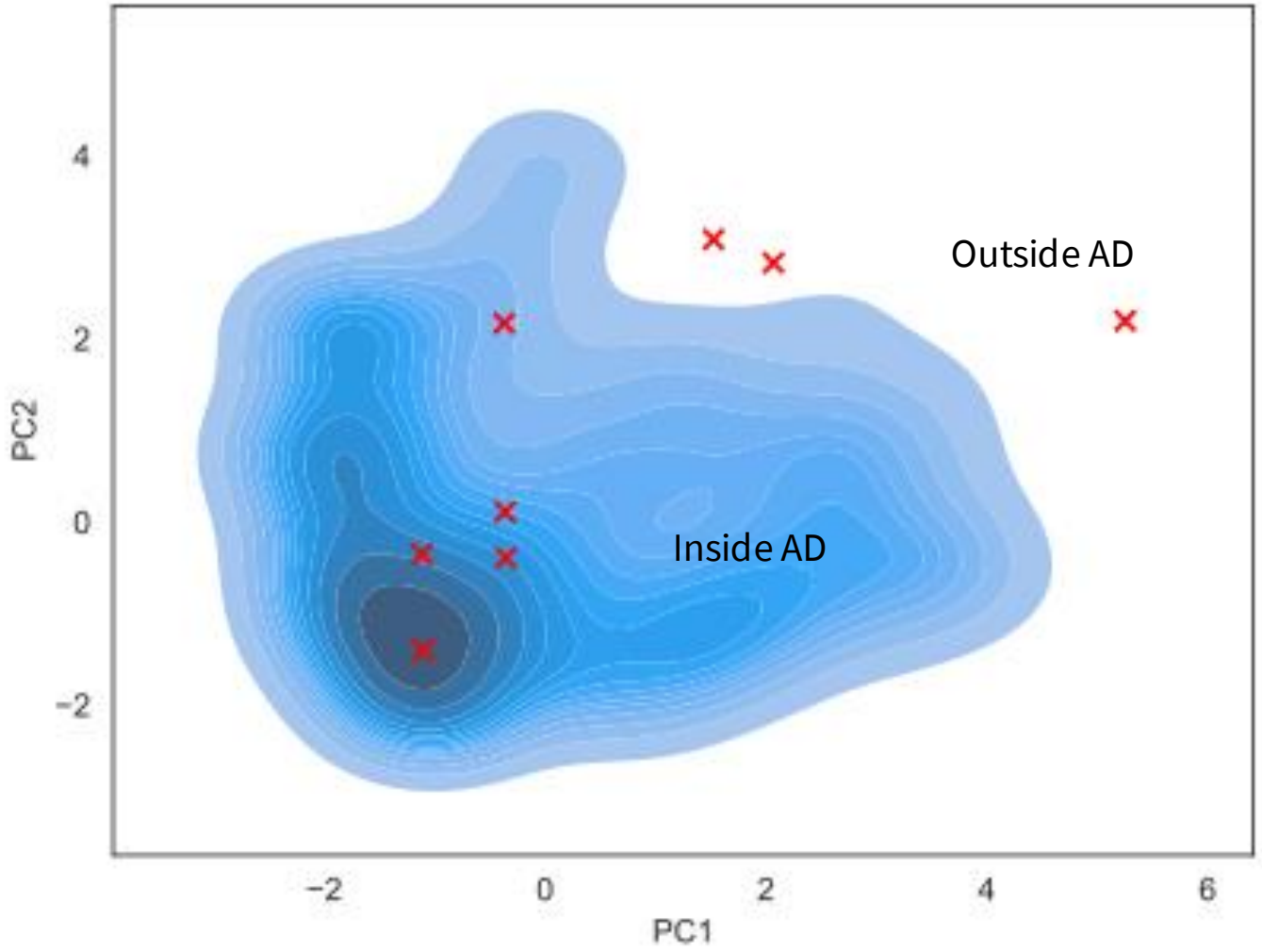
x1, x2: features 1 and 2

red dot: center of input space
the farther a point is from the center, the higher leverage

yellow ellipse: points with **same leverage** (independent of their target variable)

https://ch.mathworks.com/help/stats/hat-matrix-and-leverage.html
T.-M.Dutschmann, V.Schlenker, K.Baumann, Molecular Informatics2024, 43, e202400018. https://doi.org/10.1002/minf.202400018

AD metrics – Kernel Density Estimation (KDE)



Use KDE to estimate **training data density** in descriptor space.

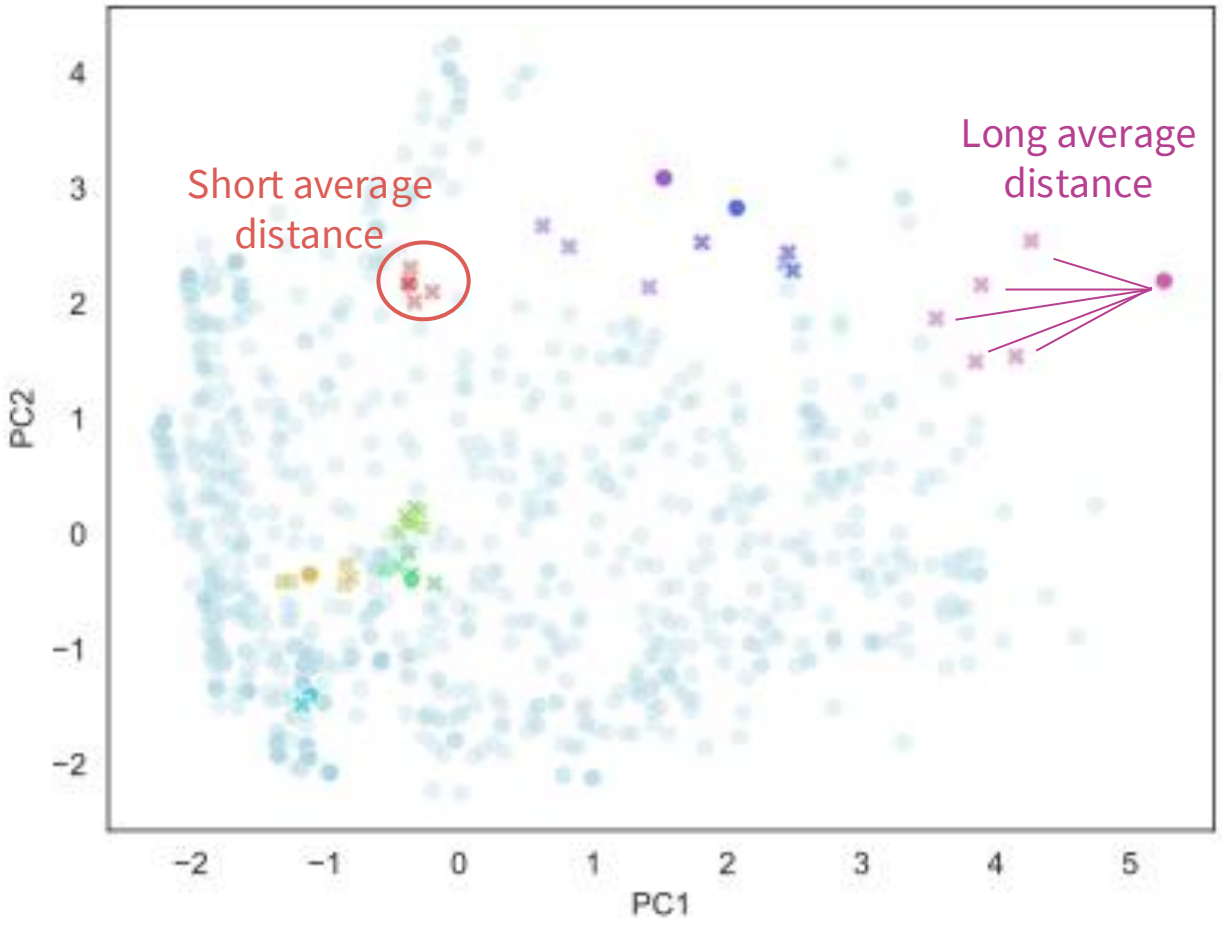
Use in combination with **dimensionality reduction**

- For **visualization**
- When descriptor space is **sparse**

T.-M.Dutschmann, V.Schlenker, K.Baumann, Molecular Informatics2024, 43, e202400018. https://doi.org/10.1002/minf.202400018

AD metrics – Distance to k-nearest neighbors

Average **distance** to the **k nearest neighbors** in training space



- **Local** metric
- **Interpretability**: retreive data points (e.g., molecules) on which the prediction is based

Query point
Closest neighbors

T.-M.Dutschmann, V.Schlenker, K.Baumann, Molecular Informatics2024, 43, e202400018. https://doi.org/10.1002/minf.202400018

AD metrics – Chemical similarity: Tanimoto distance to nearest neighbor

Tanimoto distance/similarity of **fingerprints** (equivalent to Jaccard index) is a common metric to **compare molecular structures**

Step 1: calculate pairwise similarities between all substances

Molecule A: 100100100101 → **A = 5**
Molecule B: 010100100100 → **B = 4**

In common: → **C = 3**

$$\text{Tanimoto similarity} = \frac{C}{A+B-C} = \frac{3}{5+4-3} = 0.5$$

Step 2: Identify most similar substance to query and get associated similarity → Tanimoto similarity to nearest neighbor

- Similar to kNN distance, but directly based on structural similarity
- **Local** metric
- **Interpretability**: retreive data points (e.g., molecules) on which the prediction is based

https://www.drugdesign.org/chapters/molecular-similarity/

Exercise: Applicability Domain

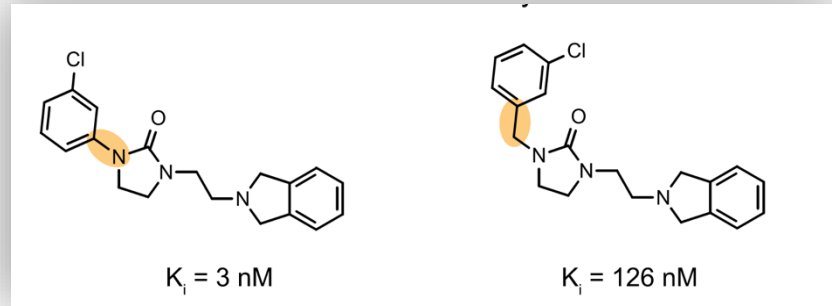
1) Calculate distance scores for the training data in the ESOL data set. Reuse calculated fingerprints/descriptors from first exercise.
   1) Distance from centroid
   2) Kernel density estimation
   3) Distance from k-nearest neighbors
   4) Tanimoto similarity
2) Define an AD threshold for each distance scores
3) Which of your test molecules are within the AD? Which ones are beyond?
4) Which of the distance scores would you choose for your descriptors, and why?

Limitation of QSARs: Activity cliffs

Key assumption: similar substances have similar properties/activities

But what if the assumption breaks? The activity landscape is not always smooth → **activity cliffs**

Example of an activity cliff:



Smooth landscape **Rough landscape**

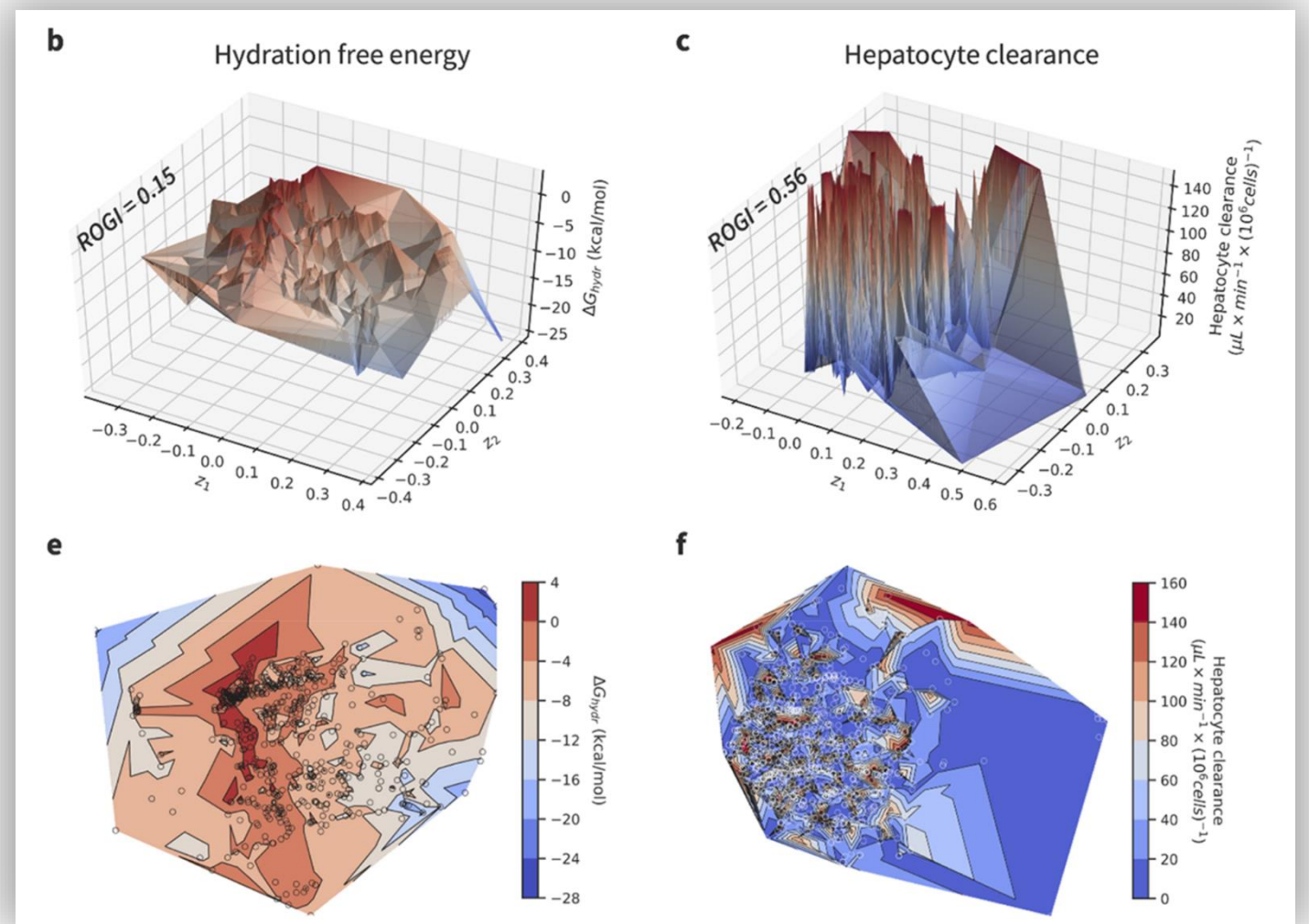


Figure left: D. van Tilborg, A. Alenicheva and F. Grisoni, Exposing the Limitations of Molecular Machine Learning with Activity Cliffs, *J. Chem. Inf. Model.*, 2022, **62**, 5938–5951.
Figure right: M. Aldeghi, D. E. Graff, N. Frey, J. A. Morrone, E. O. Pyzer-Knapp, K. E. Jordan and C. W. Coley, Roughness of Molecular Property Landscapes and Its Impact on Modellability, *J. Chem. Inf. Model.*, 2022, **62**, 4660–4671.

Take-home messages

- When you **use** someone else's **model**

  → carefully **check** if the model is applicable to your problem/substances

- When you **build** a **model** for others
  → carefully **report** when the model can be used, and when unreliable predictions are expected

- **Addition/alternative** to AD: **Uncertainty** (e.g., standard deviation) of model **predictions**

  → *More on this next Wednesday (lecture 21)*